Laboratory for Scientific Computing
UNIVERSITY OF CAMBRIDGE | Cavendish Laboratory
Department of Physics

Mathematical Methods for Scientific Computing

Matrices

Stephen Millmore

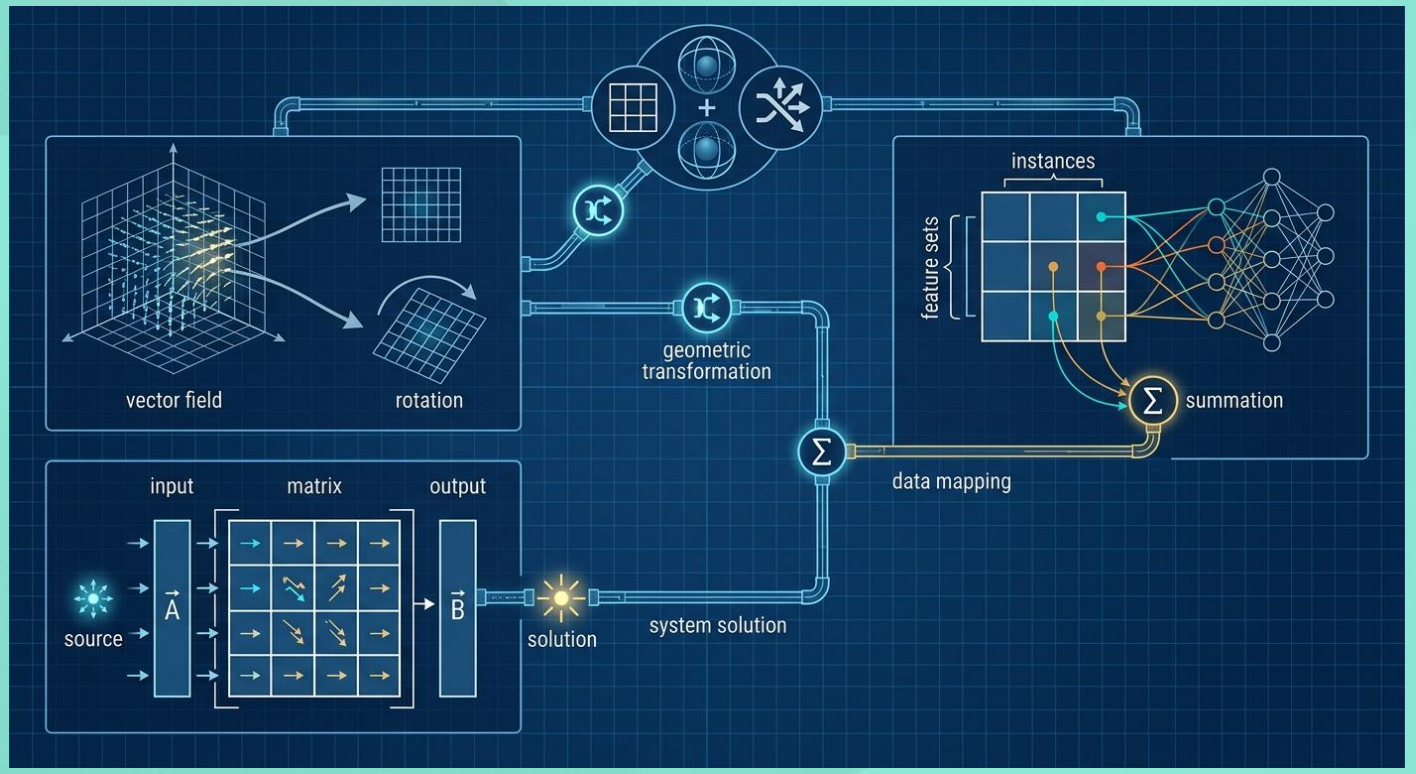
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Outline

- 1. Properties of matrices
- 2. Elementary operations
- 3. The inverse
- 4. The determinant
- 5. Eigenvalues and eigenvectors
- 6. LU decomposition

Properties of matrices



Matrices

- We briefly mentioned matrices when talking about index notation. Now we shall go into more details about their properties
- The key focus in this topic is matrices used for solving systems of equations, where it is more common to work with larger ($N \times N$) systems than the three-dimensional systems we considered under vector calculus
- The fact that such matrices are encountered throughout scientific computing is probably not a surprise
- They are used in solving linear systems of equations (e.g. incompressible flow) or analysing the properties of systems of partial differential equations
- We could argue that the only things we need to know about matrices are how to find eigenvalues (and eigenvectors) and how to compute the inverse
- In scientific computing, though, we may wish to do these things quickly, and that is not always simple

Matrices and vectors

- When understanding properties of matrices, or when converting a system of equations into matrix form, it can sometimes be convenient to think of matrices as a collection of vectors
- For example,

$$\mathrm{A} = (\mathbf{v}_1 | \mathbf{v}_2 | \mathbf{v}_3 | \cdots | \mathbf{v}_M)$$

where $\mathbf{v}_i$ is a column vector

Note – this is not true index notation here, since the vector remains a vector

- Or, if $\mathbf{w}_j$ is a row vector

$$\mathrm{B} = \begin{pmatrix} \mathbf{w}_1 \\ \mathbf{w}_2 \\ \mathbf{w}_3 \\ \vdots \\ \mathbf{w}_M \end{pmatrix}$$

Linear dependency

- Many key matrix properties are related to the idea of linear dependency in the matrix rows or columns
- Rows are **linearly dependent** if one row of the matrix can be written as a linear combination of other rows
- Mathematically, we say that a matrix is linearly independent if

$$a_1\mathbf{v}_1 + a_2\mathbf{v}_2 + a_3\mathbf{v}_3 + \cdots + a_M\mathbf{v}_M = \mathbf{0}$$

  is satisfied only if $a_i = 0$ for all i

- Otherwise, if non-zero values exist which satisfy this relationship, we could write

$$\mathbf{v}_1 = \frac{1}{a_1}(a_2\mathbf{v}_2 + a_3\mathbf{v}_3 + \cdots + a_M\mathbf{v}_M)$$

- That is, we can write one row as a combination of the other rows (or, equivalently, one column as a combination of the others)

Familiar types of matrix

Most of these properties require that the matrix is square – this is the most common type found in scientific computing

- Symmetric matrices

$$A_{ij} = A_{ji}$$

- Antisymmetric matrices

$$A_{ij} = -A_{ji}$$

- Diagonal matrices

$$A_{ij} = 0 \text{ if } i \neq j$$

- Upper triangular matrices

$$B_{ij} = \begin{cases} b_{ij} & i \leq j \\ 0 & i > j \end{cases}$$

- Lower triangular matrices

$$C_{ij} = \begin{cases} c_{ij} & i \geq j \\ 0 & i < j \end{cases}$$

Familiar square matrix properties

Note that $\mathbf{A}^1$ is a vector made from the first row of A, and A^1_k is an index notation representation of this vector

- Trace of a matrix: $\mathrm{Tr(A)} = A_{ii}$
- Inverse, A^{-1}: $(A^{-1})_{ik}A_{kj} = \delta_{ij}$
- Determinant,

$$\det(\mathrm{A}) = \sum_k (-1)^{1+k} \mathrm{M}_{1,k} A^1_k$$

  where $\mathrm{M}_{i,\mathrm{j}}$ is the **minor** of A – the matrix with row i and column j removed

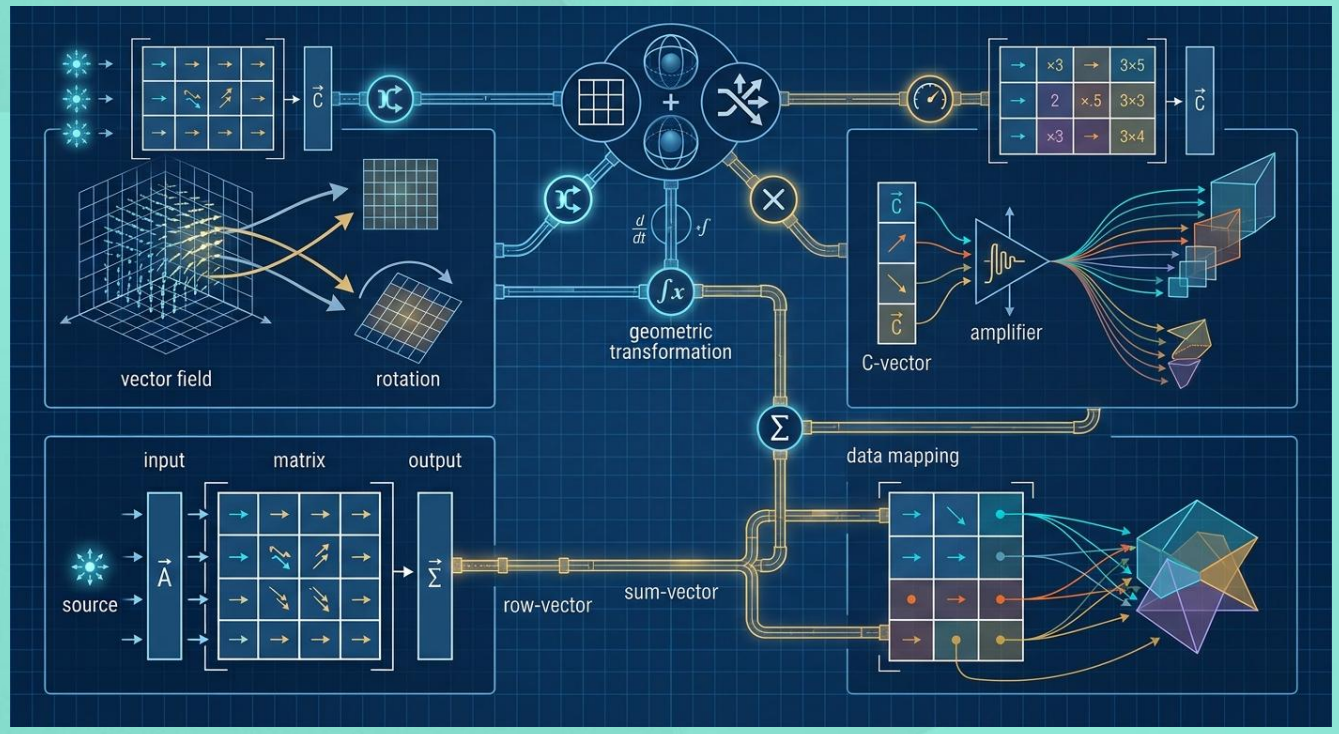
- It is hard (but not impossible), and usually impractical, to write a single line expression for the determinant in index notation

$$\det(\mathrm{A}) = \epsilon_{i_1 i_2 i_3 \ldots i_N} A^1_{i_1} A^2_{i_2} A^3_{i_3} \ldots A^N_{i_N}$$

- An extension of the Levi-Civita symbol is used here

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Elementary operations



Elementary row and column operations

- When working with matrices, we often need to move rows and/or columns about
- This might be when solving simultaneous equations, computing a determinant or inverting a matrix
- These are likely very familiar in the first application

$$\begin{aligned} ax + by + cz &= j \\ dx + ey + fz &= k \\ gx + hy + iz &= l \end{aligned}$$

$$\begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} j \\ k \\ l \end{pmatrix}$$

- When solving this system, it is likely you will use three operations:
    1. Multiplying a row by a scalar: $r_i \to \lambda r_i$
    2. Adding a multiple of one row to another: $r_i \to r_i + \mu r_j$
    3. Exchanging rows: $r_i \leftrightarrow r_j$
- These are **elementary row operations** or simply **row operations**

Elementary row (column) operation matrices

- Each of the elementary row (or column) operations can be considered as a matrix which multiplies our initial matrix, or equation, from the left (or right, for column operations)

- $r_2 \to \lambda r_2$:

$$\mathrm{E}_\lambda = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

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$$\mathrm{E}_\mu = \begin{pmatrix} 1 & 0 & \mu \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

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- $r_2 \leftrightarrow r_3$:

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- To perform column operations, we simply reverse the order of the multiplication (e.g. AE_λ instead of $\mathrm{E}_\lambda \mathrm{A}$)

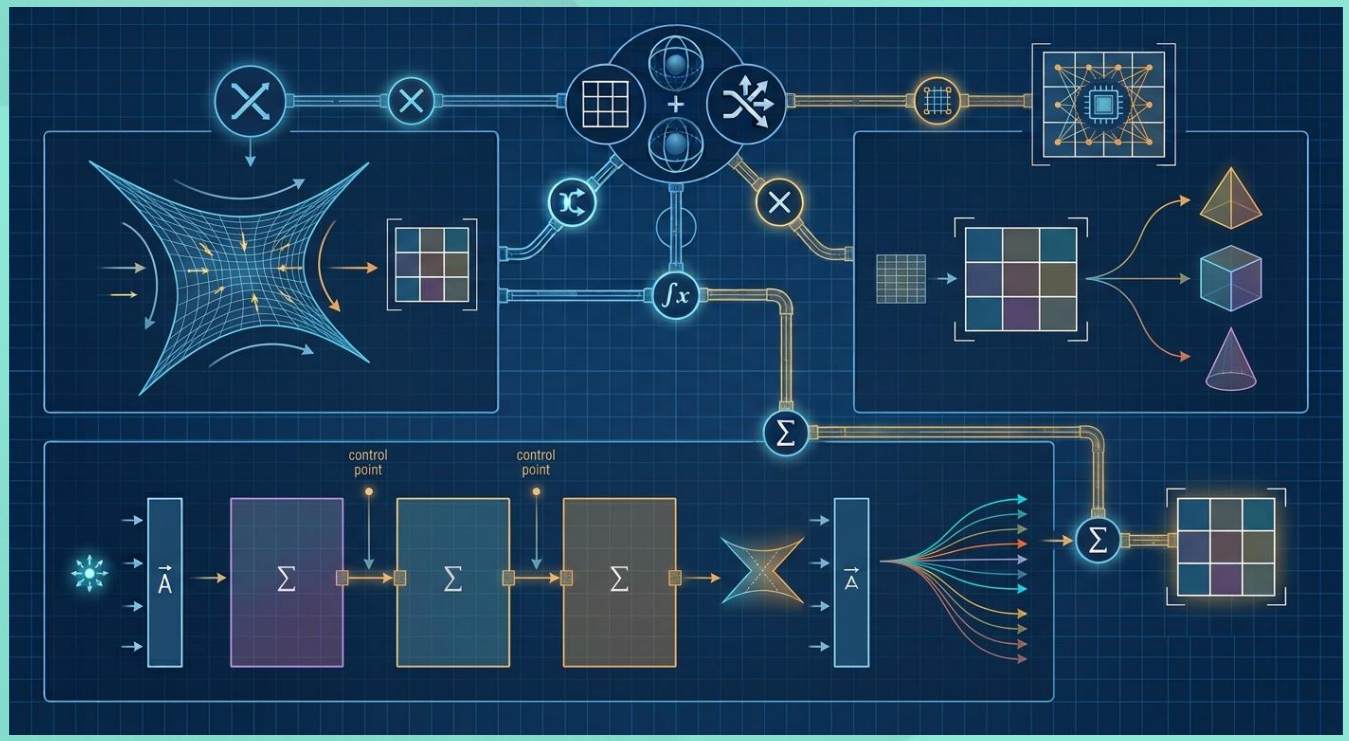
Elementary matrix notation

- Typically, multiple elementary operations are done on a single matrix, each operation denoted E_α
- This can be denoted

$$\mathrm{A}' = \mathrm{E}_n \mathrm{E}_{n-1} \dots \mathrm{E}_2 \mathrm{E}_1 \mathrm{A} = \mathrm{E}\mathrm{A}$$

- Note that E, the matrix formed by multiplying the elementary operation matrices, is not necessarily a unique combination of these operations
- Also note that the new matrix A' is related to the matrix A, we shall see some of these relations soon

The inverse



Properties of the inverse

- Commutativity: $\mathrm{A}\mathrm{A}^{-1} = I$ and $\mathrm{A}^{-1}\mathrm{A} = I$
- Inverse of an inverse: $(\mathrm{A}^{-1})^{-1} = \mathrm{A}$

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- Scalar multiplication: $(k\mathrm{A})^{-1} = \frac{1}{\mathrm{k}}\mathrm{A}^{-1}$
- Inverse of a product: $(\mathrm{AB})^{-1} = \mathrm{B}^{-1}\mathrm{A}^{-1}$
- Which extends to: $$(\mathrm{A}_1\mathrm{A}_2 \dots \mathrm{A}_n)^{-1} = (\mathrm{A}_n)^{-1} \dots (\mathrm{A}_2)^{-1}(\mathrm{A}_1)^{-1}$$

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- Which extends to: $$(\mathrm{A}_1\mathrm{A}_2 \ldots \mathrm{A}_n)^{-1} = (\mathrm{A}_n)^{-1} \ldots (\mathrm{A}_2)^{-1}(\mathrm{A}_1)^{-1}$$
- Not every matrix has an inverse, but the elementary matrices do

Inverse of the elementary matrices

- In order to show further properties of the inverse, and matrices in general, it is important to show that elementary matrices are invertible
- This is probably not too surprising – it is equivalent to row (or column) operations being reversible, i.e. we have E_1 and E_2 such that E_2 reverses the operation done by E_1

$$\mathrm{E}_2\mathrm{E}_1\mathrm{A} = \mathrm{A}$$

- Multiplication of a row by a scalar is reversed by division of a row by the same scalar
- Adding a multiple of a row is reversed by subtracting the same multiple
- Switching two rows is reversed by simply switching them again
- Because all these operations are possible, we must have

$$\mathrm{E}_2 = (\mathrm{E}_1)^{-1}$$

Non-invertible matrices

- If we consider the simultaneous equation $\mathbf{A}\mathbf{x} = \mathbf{b}$, we can write

$$\mathbf{A}^{-1}\mathbf{A}\mathbf{x} = \mathbf{x} = \mathbf{A}^{-1}\mathbf{b}$$

- This implies that there is a unique solution for $\mathbf{x}$
- We know this isn't true for all simultaneous equations, some cannot be solved, whilst others have infinite solutions

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- We know this isn't true for all simultaneous equations, some cannot be solved, whilst others have infinite solutions
- In that case, A^{-1} does not exist
- A system of simultaneous equations does not give a unique solution when one of the equations can be written as a combination of the others – when there is a **linear dependency** between the equations
- We can show that linearly dependent matrices do not have an inverse

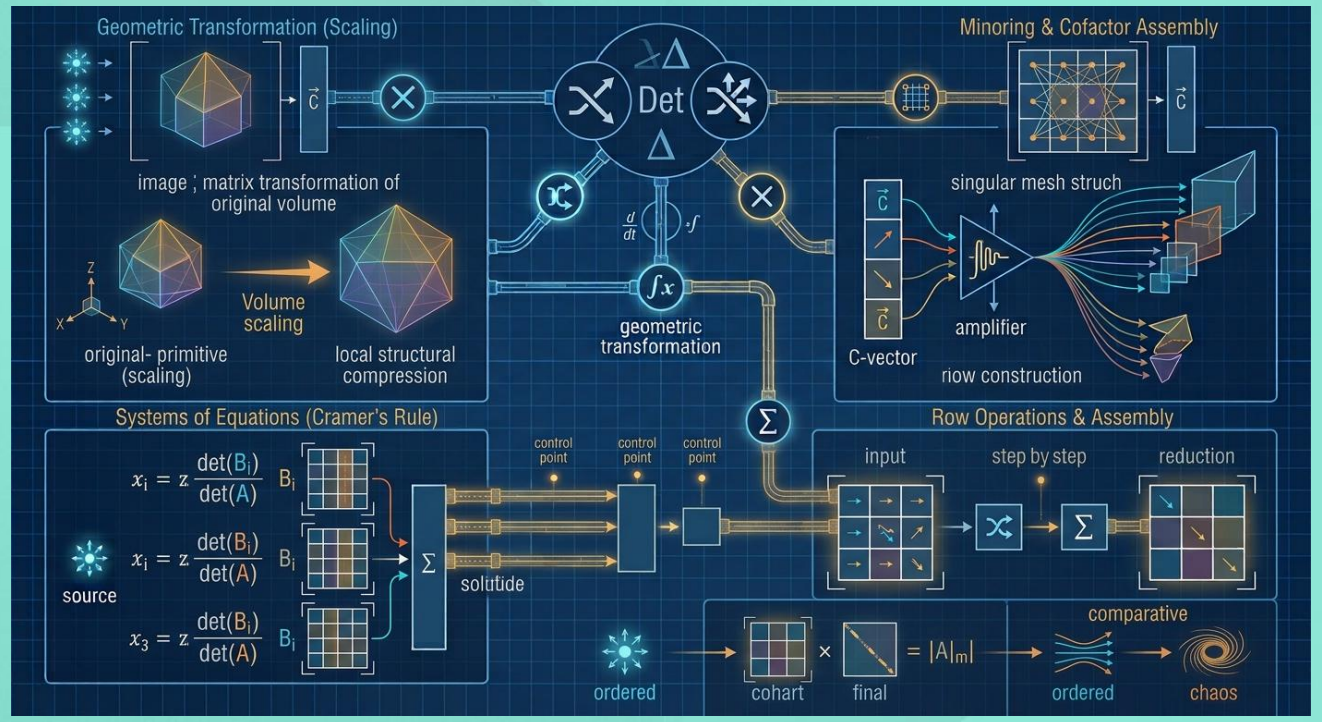
Invertible matrices

- The converse to the previous slide is also true – linearly independent matrices have an inverse
- We can show that for a linearly independent matrix, a series of row operations exist such that

$$\mathrm{EA} = I$$

- This is done by first putting our matrix A in **row echelon form**, and when we do this, we know that all diagonal entries must be non-zero
- In doing so, we find that $\mathrm{A}^{-1} = \mathrm{E}$
- This then also gives us a technique to find the inverse

The determinant



Determinants of diagonal matrices

- The determinant itself is not too tricky to calculate, at least when we have a matrix that is just numbers (and we don't make typos)
- Scientific computing, however, often requires determinants of algebraic matrices, and additionally, may need to simplify the algebraic expression such that we understand physical or mathematical properties of our system
- If we are lucky enough to have a diagonal matrix, or an upper or lower triangular matrix, then the determinant is straightforward to compute – simply multiply the diagonal entries together, and generally easier to simplify
- We know we can convert a matrix to row echelon form, and from that, the determinant is now easy to compute, but in using a series of row operations, we might not recover the determinant of the original matrix
- All we need to know is what effect the elementary operations we used to obtain row echelon form have on the determinant
- Fortunately, this is easy to work out

Effect of elementary operations on determinants

- Multiplying a row by a scalar multiplies the determinant by the same scalar

$$A' = E_\lambda A \quad \Longrightarrow \quad \det(A') = \lambda \det(A)$$

Effect of elementary operations on determinants

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$$B' = E_\mu B \implies \det(B') = \det(B)$$

Effect of elementary operations on determinants

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$$B' = E_\mu B \implies \det(B') = \det(B)$$

- Switching two rows switches the sign of the determinant

$$C' = E_\nu A \implies \det(C') = -\det(C)$$

Effect of elementary operations on determinants

- Multiplying a row by a scalar multiplies the determinant by the same scalar

$$\mathrm{A}' = \mathrm{E}_\lambda \mathrm{A} \quad \Longrightarrow \quad \det(\mathrm{A}') = \lambda \det(\mathrm{A})$$

- Adding a multiple of one row to another doesn't change the determinant

$$\mathrm{B}' = \mathrm{E}_\mu \mathrm{B} \Longrightarrow \det(\mathrm{B}') = \det(\mathrm{B})$$

- Switching two rows switches the sign of the determinant

$$\mathrm{C}' = \mathrm{E}_\nu \mathrm{A} \Longrightarrow \det(\mathrm{C}') = -\det(\mathrm{C})$$

- This means multiple elementary operations either multiply the determinant by a scalar (including -1) or leave it unchanged
- For an arbitrary number of operations, E, we have

$$\mathrm{A}' = \mathrm{E}\mathrm{A} \Longrightarrow \det(\mathrm{A}') = \alpha \det(\mathrm{A})$$

Determinants and linear dependency

- If a matrix has a row which is entirely zeroes, it is easy to see (and show) that the determinant is zero
- For a linearly dependent matrix, we can perform elementary operations,

$$\mathrm{A}' = \mathrm{EA}$$

  such that A' contains a row of zeroes

- Therefore

$$\det(\mathrm{A}') = 0 \quad \text{and} \quad \det(\mathrm{A}') = \alpha \det(\mathrm{A})$$

- This means $\det(A) = 0$ if A has linearly dependent rows or columns
- The converse also holds – a linearly independent matrix, B, has $\det(B) \neq 0$
- As a result, the determinant is a quick method to check for linear dependency

Determinant of a product

- The determinant of a matrix product obeys

$$\det(\mathrm{AB}) = \det(A)\det(B)$$

- Although this appears to be a simple expression, it is not a quick proof
- We consider the cases of A being linearly dependent and independent separately
- **Linear dependency**: since $\det(AB) \neq 0$ would imply that $(\mathrm{AB})^{-1}$ exists, we should have

$$(\mathrm{AB})(\mathrm{AB})^{-1} = \mathrm{ABB}^{-1}\mathrm{A}^{-1} = I$$

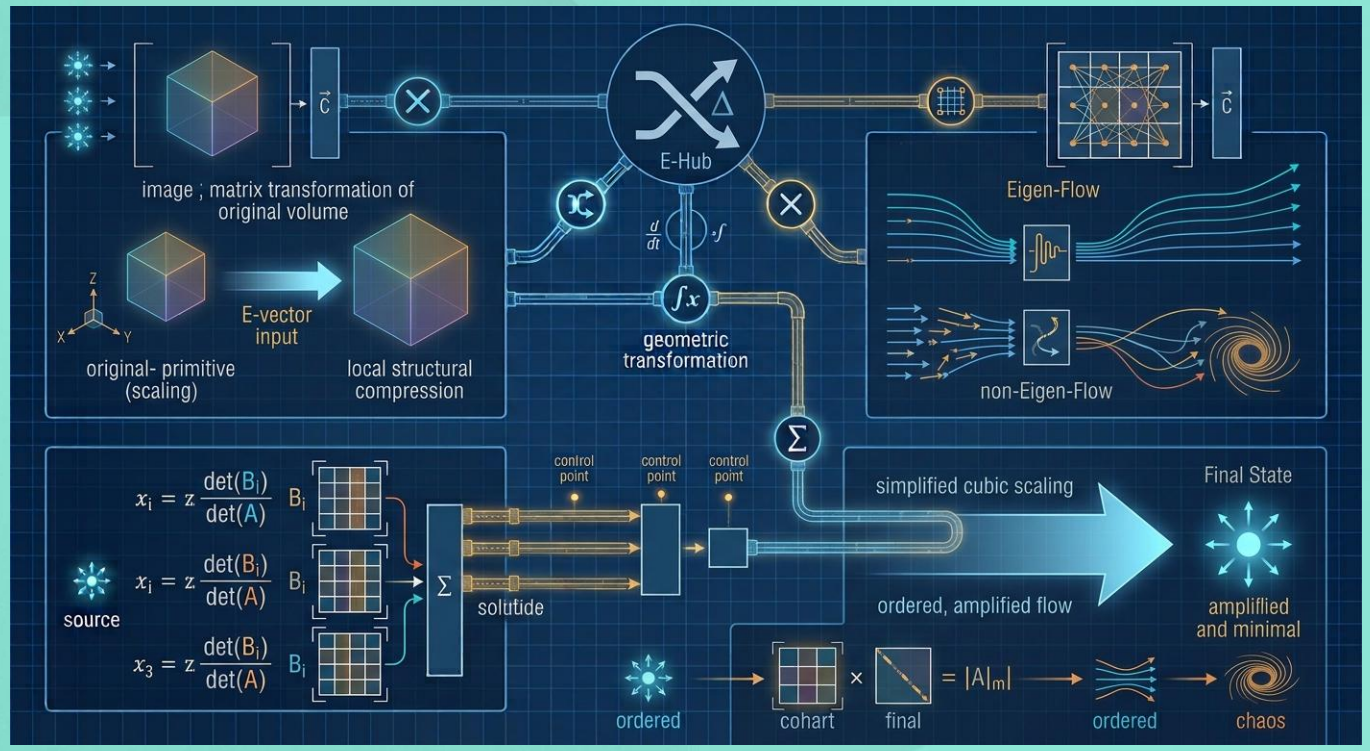
  but since A^{-1} doesn't exist, this is a contradiction

- **Linear independence**: in this case, we know we have $\mathrm{A} = \mathrm{E}I$, so

$$\det(\mathrm{AB}) = \det(\mathrm{EB}) = \alpha\det(\mathrm{B})$$

- By choosing the case of $\mathrm{B} = I$, we see $\alpha = \det(\mathrm{A})$

Eigenvalues and eigenvectors



Matrices as coordinate transformations

- A related concept to eigenvalues and eigenvectors is the fact that a vector multiplied by a matrix is a linear map of that vector to a new coordinate system
- We often don't need to consider this explicitly, but sometimes it is useful
- For example, if we have $\mathbf{v}' = \mathrm{E}_\nu \mathbf{v}$, with,

$$\mathrm{E}_\nu = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

  we change the handedness of the coordinate system

- And for $\mathbf{v}' = \mathrm{R}\mathbf{v}$ with

$$\mathrm{R} = \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

  we rotate our coordinate system (around the z-axis)

Eigenvalues and eigenvectors

- Sometimes, multiplication of a vector by a matrix will not change the vector, or will simply scale it (but not change its direction)

$$\begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 3 \end{pmatrix}$$

- This can be very useful for making absolute statements about a system
- And this is behind the definition of eigenvalues and eigenvectors

$$\mathrm{A}\mathbf{v} = \lambda\mathbf{v}$$

- Or we could have

$$\mathbf{w}^T\mathrm{A} = \lambda\mathbf{w}^T$$

which are the **left eigenvectors** of the system

- $\mathbf{v}$ in the expression above is a **right eigenvector**, or often, simply an eigenvector

Finding eigenvalues

$$\mathrm{A}\mathbf{v} = \lambda\mathbf{v}$$

- We want to find non-zero solutions to

$$(\mathrm{A} - \lambda I)\mathbf{v} = \mathbf{0}$$

- The zero vector, $\mathbf{0}$, is always trivially a solution – this is not an eigenvector
- The only way we can obtain non-zero solutions is if $(\mathrm{A} - \lambda I)$ is a linearly dependent matrix
- This gives us an easy way to find the eigenvalues $\det(\mathrm{A} - \lambda I) = 0$
- We get a polynomial in λ which is satisfied by the eigenvalues – this also tells us

$$\det(\mathrm{A}) = \lambda_1\lambda_2\lambda_3 \cdots \lambda_N$$

Finding eigenvectors

- Once you have your eigenvalues, finding the eigenvectors is a matter of solving a simultaneous equation

$$\mathrm{A} = \begin{pmatrix} 0 & -4 & -3 \\ -3 & 1 & 3 \\ -2 & 1 & 4 \end{pmatrix}$$

- This matrix has eigenvalues $\lambda_1 = 1$, $\lambda_2 = -3$, $\lambda_3 = 7$
- Therefore one eigenvector is obtained by solving $(\mathrm{A} - 7I)\mathbf{v} = 0$,

$$\begin{pmatrix} -7 & -4 & -3 \\ -3 & -6 & 3 \\ -2 & 1 & -3 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \mathbf{0}$$

- We know that there is not a unique solution to this equation, and what we find is

$$\mathbf{v}_3 = a \begin{pmatrix} 1 \\ -1 \\ -1 \end{pmatrix}$$

for any constant a – it is common to choose a such that $\mathbf{v}$ is normalised

Finding eigenvalues – other possibilities

- Our first example considered three unique real eigenvalues
- There is no need for them to be either of these – complex eigenvalues are common, and result in complex eigenvectors, and repeated eigenvalues are common too (see next slide)
- The eigenvalues of diagonal and triangular matrices are particularly easy – they are the diagonal entries
- Elementary row operations do not change the eigenvalues of a matrix – this means you can sometimes manipulate a matrix to make finding the eigenvalues easier
- Linearly dependent matrices always have at least one eigenvalue of zero

Finding eigenvectors – repeated eigenvalues

- We might find we have as many eigenvectors as we have unique eigenvalues

$$\mathrm{A} = \begin{pmatrix} 5 & -2 & 0 \\ 0 & 4 & 0 \\ 3 & -5 & 5 \end{pmatrix}$$

- This matrix has eigenvalues $\lambda_1 = 4$, $\lambda_{2,3} = 5$
- Here we find two eigenvectors,

$$\lambda_1 \colon \hat{\mathbf{v}}_1 = \frac{1}{\sqrt{6}} \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix} \qquad \lambda_2 \colon \hat{\mathbf{v}}_2 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

Finding eigenvectors – repeated eigenvalues

- We might find we have more eigenvectors than we have unique eigenvalues

$$\mathrm{A} = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$$

- This matrix has eigenvalues $\lambda_1 = 2$, $\lambda_{2,3} = -1$
- Here we could say that we find two eigenvalues,

$$\lambda_1 \colon \hat{\mathbf{v}}_1 = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \qquad \lambda_2 \colon \hat{\mathbf{v}}_2 = \begin{pmatrix} x \\ y \\ -x - y \end{pmatrix}$$

- However, we could split $\hat{\mathbf{v}}_2$ into two vectors (one using x, one using y):

$$\hat{\mathbf{v}}_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}, \qquad \hat{\mathbf{v}}_3 = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}$$

Linear independence of eigenvectors

- Whilst it is clear that all the eigenvectors we have found so far are different, this doesn't necessarily mean they are linearly independent
- However, it is possible to show that they are – the eigenvectors from distinct eigenvalues are always linearly independent
- For cases where repeated eigenvalues can be written as more than one eigenvector, we can also **choose** these to be linearly independent
- A result of this is that two eigenvalues can never have the same eigenvector
- In fields where eigenvectors form a useful coordinate system, knowing that they are independent is particularly important

Diagonalisation

- Once we have identified the eigenvalues of a matrix, we know that a diagonal matrix, with these eigenvalues as the entries, would clearly have the same eigenvalues
- This means there is a technique, **diagonalisation** which allows you to connect these two matrices
- Specifically, we have

$$\mathrm{A} = \mathrm{S}\Lambda\mathrm{S}^{-1}$$

  where S is the matrix whose columns are the eigenvectors of A

- Not all matrices are diagonalisable, e.g. an $N \times N$ matrix without N eigenvectors
- But where a matrix is diagonalisable, this has useful implications, including when computing matrix powers

$$\mathrm{A}^m = \mathrm{S}\Lambda^m\mathrm{S}^{-1}$$

Similar matrices

- The relationship we saw for diagonalisable matrices does generalise – any matrices A and C related by

$$\mathrm{C} = \mathrm{BAB}^{-1}$$

  are **similar matrices**

- Similar matrices have some useful properties:
    1. They have the same determinant

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- Similar matrices have some useful properties:
  1. They have the same determinant
  2. They have the same eigenvalues
  3. They have related eigenvectors (but not the same eigenvectors)
- When dealing with algebraic matrices in particular, similar matrices can be useful to make it easier to compute some of these properties

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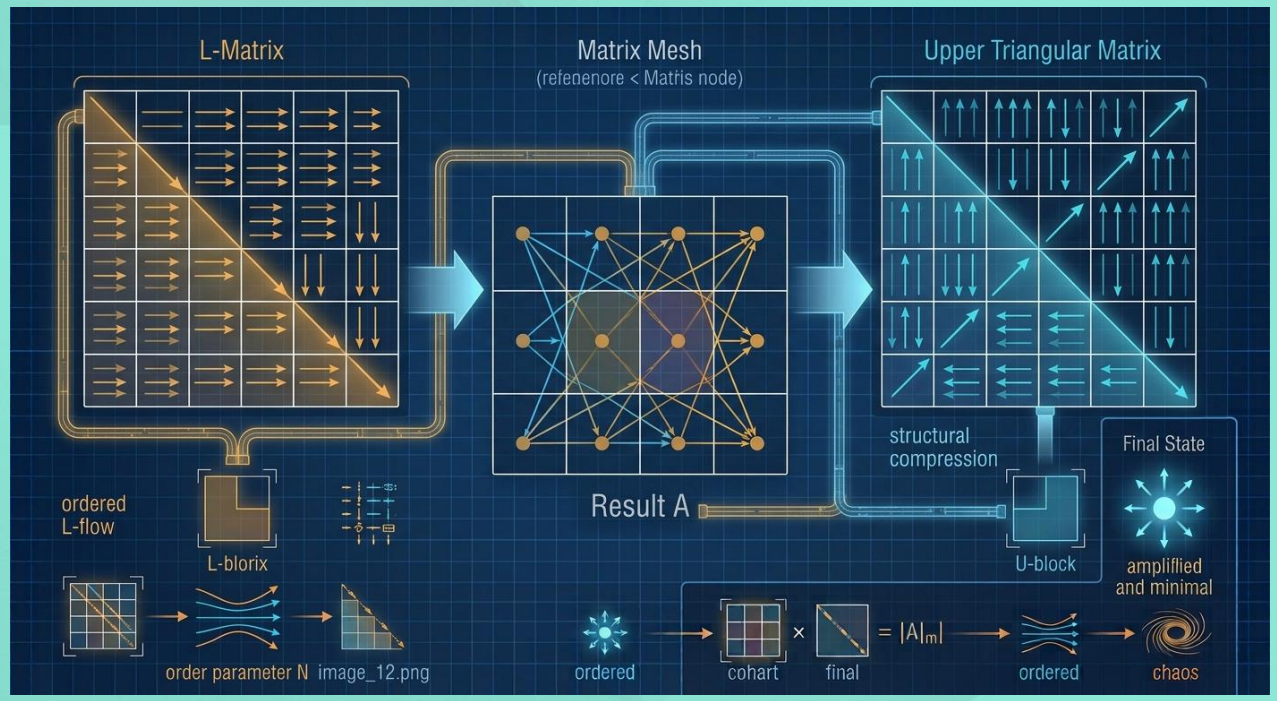
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  2. They have the same eigenvalues
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Symmetric matrices

- It is quite common for physical systems with matrix quantities to have symmetric matrices
- These have particularly useful properties which can greatly simplify computations
- The eigenvalues of a symmetric matrix are **always real** – this is very useful to know if you want to compute these computationally
- The eigenvectors of a symmetric matrix are always orthogonal – we can sometimes use this to create a coordinate system with useful physical properties

LU decomposition



Triangular matrices in simultaneous equations

- We know that we can use various matrix operations to solve simultaneous equations

$$\mathbf{A}\mathbf{x} = \mathbf{b}$$

- However, in scientific computing, A could be large – algorithms which work through row reduction operations can be very expensive
- Sometimes A has nice properties, however

$$\begin{pmatrix} 1 & 1 & 1 \\ 0 & 2 & 1 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}$$

- Here, we can read off the solution for z, and use this directly to find y, and then use that to find x (and similar for higher dimensions)
- Lower triangular matrices have similar nice properties

LU decomposition

- For any matrix, we can perform elementary operations to obtain,

$$\mathrm{A} = \mathrm{EU}$$

  where U is an upper-triangular matrix

- **Now**, if none of the elementary operations were row-switching operations, then we have,

$$\mathrm{E} = \mathrm{L}$$

  a lower-triangular matrix

- This is the LU decomposition, we can write

$$\mathrm{A} = \mathrm{LU}$$

- Solving the simultaneous equations is now straightforward

Using LU decomposition

- From $\mathrm{A}\mathbf{x} = \mathbf{b}$ we have;

$$\mathrm{LU}\mathbf{x} = \mathbf{b}$$

- Now set $\mathbf{y} = \mathrm{U}\mathbf{x}$,

$$\mathrm{L}\mathbf{y} = \mathbf{b}$$

- This is a lower-triangular simultaneous equation, it is easy to solve for $\mathbf{y}$
- But now, $\mathbf{y} = \mathrm{U}\mathbf{x}$ is an upper-triangular simultaneous equation with known $\mathbf{y}$ – this is also easy to solve
- So, provided L and U were constructed with the right properties, instead of row-reductions on A directly, we did row reductions on A to get U (and L at the same time), then use this to solve the equation
- For a single equation, this hasn't gained us a lot, keeping track of two matrices is definitely harder than row-reducing one, but…

Applications of LU decomposition

- It is common to need to solve $\mathrm{A}\mathbf{x} = \mathbf{b}$ repeatedly for different values of $\mathbf{x}$ but a single A
- Or, maybe more accurately, for a matrix A which has algebraic entries, so performing the LU decomposition once is time-consuming, but then each time it is used, you just plug in numbers directly into the lower- and upper-triangular matrices
- LU decomposition means we only perform the complex elementary operations once, rather than for every equation – a large computational saving
- However, we said that we must avoid row switching for this to work; the chance our elementary operations avoid row switching is quite small
- In that case, we first work out what switches are needed to obtain U, perform these on the system before decomposing, and then compute L and U for the newly ordered system – again, this could be time consuming, but it only needs doing **once**